

Multiple Choice (10 marks)

1. In the diamond structure of elemental carbon, the nearest neighbors of each C atom lie at the corners of a
 - (a) square
 - (b) hexagon
 - (c) cube
 - (d) tetrahedron
2. Which of the following statements about bosons and/or fermions is true?
 - (a) Bosons have symmetric wave functions and obey the Pauli exclusion principle.
 - (b) Bosons have antisymmetric wave functions and do not obey the Pauli exclusion principle.
 - (c) Fermions have symmetric wave functions and obey the Pauli exclusion principle.
 - (d) Fermions have antisymmetric wave functions and obey the Pauli exclusion principle.
3. Which of the following is true about any system that undergoes a reversible thermodynamic process?
 - (a) There are no changes in the internal energy of the system.
 - (b) The temperature of the system remains constant during the process.
 - (c) The entropy of the system and its environment remains unchanged.
 - (d) The entropy of the system and its environment must increase.
4. By definition, the electric displacement current through a surface S is proportional to the
 - (a) rate of change of the electric flux through S
 - (b) electric flux through S
 - (c) time integral of the magnetic flux through S
 - (d) rate of change of the magnetic flux through S
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True / False (10 marks)

Answer True or False

6. True or False: In the observer's reference frame, it takes 2.5 ns for a meter stick moving at $0.8c$ to pass the observer.
7. True or False: The angular magnification of the telescope is 5, which is calculated using the separation distance of the lenses.
8. True or False: An electron in the $l = 2$ state can have only three allowed values of the magnetic quantum number m_l .
9. True or False: The sign of the charge carriers in a doped semiconductor can be determined by measuring the Hall coefficient.
10. True or False: According to BCS theory, the attraction between Cooper pairs in a superconductor is due to the weak nuclear force.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the conditions under which a beam of electrons can diffract when interacting with a crystal surface, and analyze the approximate kinetic energy required to observe the resulting scattering pattern.
12. Discuss the process and calculations involved in determining the final voltage across a charged capacitor when two initially uncharged capacitors are connected in series to it.

End of Paper